



Artificial intelligence and data centres

Regulatory data gaps that limit policy oversight and independent verification

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Australia already collects substantial regulatory data about data centres. **But the public records cannot yet be reliably linked at facility level, used to show the national coverage of emerging obligations, or compared on a consistent basis.** These gaps limit both policy oversight and the independent testing of electricity and water forecasts.

The recommendations below build principally on records and reporting channels that already exist, while adding limited standardisation and, where necessary, site-level water fields. None of them requires the Committee to accept any forecast, or to decide how strict any standard should be.

Three recommendations

1 Identify the facility

Establish, or publicly confirm, a persistent facility identifier across the National Greenhouse and Energy Reporting scheme (NGER), connection applications, planning, environmental referrals and relevant water records, and publish a crosswalk of non-sensitive fields. The identifier and crosswalk should remain publicly available for a minimum period.

2 Show who regulates it

Publish a jurisdictional coverage map for NEM-connected, non-NEM-connected and off-grid or self-supplied data centres, identifying the responsible regulator and the applicable obligations in each case.

3 Measure consistently

Define the reporting boundary, then require site-level reporting above a defined threshold: withdrawal, consumptive use, discharge, source type and water usage effectiveness (WUE) — not WUE alone. Capacity and efficiency figures should state their boundary and denominator.

Why the public record does not yet support this

- **Records about one facility cannot reliably be linked.** An ownership change moves a whole row in the regulator's published data though nothing was built or closed; and planning registers often do not describe the use as a data centre. (Section 2; Annex C.)
- **The proposed obligations do not yet set out a pathway for every supply configuration.** The accompanying network reform applies only in the NEM, and the obligations themselves are described principally for grid-connected facilities. Two large Northern Territory proposals show the uncertainty: Project Ares is expressly off-grid; Beetaloo Digital is proposed as self-supplied from on-site generation and is designed not to draw electricity from the existing Territory grid. (Section 3; Annex B.)
- **Water figures rest on incompatible definitions.** Published Australian WUE figures describe different levels — a facility, a region, a national fleet — and one company's FY25 reports label the same 773 ML both "consumed" and "withdrawal". (Section 4; Annex D.)

Declaration of interests. Helio Vulcan provides industrial-asset analysis to owners and investors in Australian energy projects. Some of those projects may be affected by the frameworks discussed here, so we have a commercial interest in Australian industrial energy development. We take no position on whether this industry should grow or on how strict any standard should be. Full declaration at section 1.

1. Purpose, scope and interests

This submission addresses a narrow question under term of reference (a): whether Australia's public regulatory records make it possible to identify the same data centre across government systems, determine which obligations apply to it, and compare its electricity, capacity and water data on a consistent basis.

Our review finds that the public record does not yet support those three tasks. Relevant information is already collected through energy reporting, connection, planning, environmental and water systems, but the records cannot be reliably linked at facility level. Regulatory coverage also varies by jurisdiction and supply configuration, while published capacity and water figures use inconsistent boundaries and definitions.

These findings lead to three recommendations: establish a persistent facility identifier, publish a national map of regulatory coverage, and standardise the boundaries and fields used for capacity and water reporting. Together, these measures would make the public record more usable for policy oversight and would improve the independent testing of future electricity and water forecasts.

Scope

Selecting a preferred forecast is outside the scope of this submission. The estimates before the Committee vary several-fold because they use different facility populations, project-realisation assumptions, utilisation rates, reporting boundaries and technology assumptions. Section 5 and Annex E compare those assumptions only to show why the estimates cannot yet be tested on a consistent basis; they do not endorse or reject any particular forecast.

The submission takes no position on whether the data centre industry should expand, what level should be set for an efficiency, water or emissions standard, or which forecast is most accurate. Its focus is limited to whether the public data needed to assess those questions is identifiable, comparable and sufficiently complete. Agencies may hold non-public data that we cannot see.

Relationship to other submissions

Professor Michael Vardon argues that data centre energy and water use is included within broader official statistics but cannot be separated at facility, operator or catchment level, and proposes a fix in the statistical and national accounts system.^[28] **This submission addresses a different layer: not how activity is classified in statistics, but the absence of a key that would let regulatory records be linked.** The two are complementary.

Declaration of interests

Heliovolcan is an independent industrial intelligence practice. We rebuild Australian industrial assets from public records and sell screening and analysis to owners, developers and investors in industrial energy assets, some of which would be affected by the frameworks discussed here. **We have a commercial interest in Australian industrial energy development. No commercial entity discussed as a case study in this submission is, or has been, a client of Heliovolcan.** That interest does not alter the submission's limited focus on the structure and accessibility of public regulatory data: the recommendations are limited to how existing systems connect and how records are structured, and none changes how strict any standard is.

2. Gap A — no reliable way to link records for the same facility

Any facility-level assessment must begin by establishing which records relate to the same physical site. At present the public record provides no reliable way to do this across energy reporting, connection, planning, environmental and water systems.

Each system collects information for a legitimate but different regulatory purpose, and each uses its own reporting unit and identifier. NGER has facility and corporate reporting records; EPBC referrals use referral numbers; AEMO and the network businesses use connection identifiers; and each planning jurisdiction has its own application numbers. **These identifiers are not publicly linked to one another.** Annex A describes what each system measures and the thresholds at which it applies.

NGER has a facility register. The EPBC Act has referral numbers, such as 2026/10560. AEMO has connection applications. Each State's planning system has its own application numbers. Water supply arrangements sit elsewhere again. (What each instrument measures, at what threshold: Annex A.)

We found no public crosswalk or common identifier that allows a member of the public to determine reliably whether records in these systems relate to the same physical facility. **Agencies may be able to make those matches internally** using company identifiers, addresses or non-public data-sharing arrangements, but no equivalent facility-level link is available in the public record. Our recommendation is therefore to establish or publicly confirm such an identifier, not to assert that none exists.

Two structural problems explain the failure.

- **Facility and owner names change over time, but the public records do not provide a persistent identifier that stays with the physical facility.** One Sydney facility changed hands in December 2024. In the regulator's published data the former company's row disappears (78,961 t) and the new company's row appears (77,446 t). The facility itself neither opened nor closed, but a company-level year-on-year comparison records an apparent decrease for the former owner and an apparent increase for the new owner.^[6]
- **Planning records often do not classify the use as a data centre.** A 150 MW Western Australian facility was lodged with the Development Assessment Panel as *Proposed Warehouses and Ancillary Structures*.^[27] Separately, a full-text search of 1,703 Economic Development Queensland determinations returned no occurrence of "data cent".^[24] Keyword searches therefore cannot reliably identify projects whose planning classification uses another form of development. **This strengthens the case for a persistent identifier rather than better keyword searching.** (Method and per-jurisdiction results: Annex C.)
- **Federal classification fields are unreliable too.** Project Ares — a proposed Northern Territory facility that includes 900–1,100 MW of gas-fired generation (section 3) — is recorded on the *Environment Protection and Biodiversity Conservation Act 1999* (EPBC Act) portal under the industry type "Energy Generation and Supply (renewable)", sub-type "Solar Farm".^[7] **This example shows that federal classification fields may not describe all material elements of a proposed facility, including its generation technology.**

3. Gap B — coverage depends on jurisdiction and supply configuration

The emerging NEM-based data centre obligations do not by themselves create a complete national coverage framework.

The AEMC's advice to Energy and Climate Change Ministers, *Data centre energy obligation* (July 2026), states its own scope at p 17: the policies "could be applied to all types of data centres **connected to the grid**", and the accompanying network cost-recovery reform "would **only apply to the NEM**". We also searched the 51-page detailed policy advice and found no reference to off-grid, islanded or unconnected facilities.^[4] The parallel Package 2 draft determination writes a 30 MW threshold into the National Electricity Rules,^[5] which apply to Queensland, New South Wales, the ACT, Victoria, South Australia and Tasmania. **The Northern Territory and Western Australia are not in the NEM.** (Quotations and search method: Annex B.)

Two large Northern Territory proposals illustrate why geographic location and supply configuration need to be considered separately.

Project Ares (EPBC referral 2026/10560, valid 29 June 2026)^[7] is proposed at Murrarji Station in the Barkly region. Its published materials state 1 GW of IT capacity, 3,000 MWp of solar generation, 16 GWh of battery storage and 900–1,100 MW of gas-fired generation. The referral expressly states that the facility would operate off-grid and draw no electricity from the Territory network.

Beetaloo Digital is a separate proposal at Weddell, south of Darwin. Published Northern Territory Government information describes on-site gas-fired generation of up to 2 GW supporting up to two hyperscale data centre campuses, and states that the project is being designed to add new generation capacity to the Territory rather than to draw from the existing grid.^[30]

Both proposals are outside the NEM, so the proposed NEM access standards would not apply. Their treatment under the proposed additionality and firm-capacity obligations is less clear, because the implementation route and final geographic scope of those obligations have not been settled. Project Ares also presents a distinct off-grid case; Beetaloo Digital is better described as self-supplied, because it is presented as adding generation capacity to the Territory rather than as being physically separate from every network. (Published characteristics of both proposals: Annex B.4.)

These examples do not suggest that facilities outside the NEM or operating off-grid are unregulated. National reporting law, the Safeguard Mechanism, planning, environmental, water, Territory electricity and contractual requirements may still apply. **The narrower issue is that the published instruments do not yet provide a complete and publicly checkable account of how the proposed data-centre-specific energy measures would apply outside the NEM, to entirely off-grid facilities, or to self-supplied facilities that may add generation to a network without drawing from it.**^{[4] [5]}

A published coverage map would distinguish between obligations that apply nationally, obligations that depend on NEM connection, and matters left to State or Territory law.

4. Gap C — water: boundaries before volumes

Published Australian water figures are reported at different geographic and organisational levels and cannot be compared without a common facility boundary and denominator. Requiring more disclosure does not fix that; defining the boundary does.

The first issue is whether published water figures measure the same thing. Existing figures use different geographic boundaries, organisational boundaries and definitions of water use:

- **Figures reported at different organisational and geographic levels.** Published Australian WUE figures include values for a regional fleet, a national fleet and an attributed industry average.^{[9] [12] [13]} **Each may be valid for the level at which it is reported, but the figures cannot be compared directly without a common facility boundary and denominator.** (Each figure, labelled by level: Annex D.2.)
- **The same quantity published under two definitions.** One company's FY25 sustainability reports, published together, record **773 ML** as water "consumed" across its portfolio in one report and the identical 773 ML under the heading "**Water withdrawal**" in another.^{[10] [11]} At evaporatively cooled facilities withdrawal and consumption differ materially, so the figure cannot reliably be interpreted as either withdrawal or consumption until the reporting basis is clarified. **The discrepancy appears to arise from inconsistent terminology within otherwise detailed voluntary reporting.** It illustrates why additional disclosure will not produce comparable data unless the reporting definitions are also standardised. (Exact wording, document and page: Annex D.1.)

The regional figures at Annex D.3 vary across locations, but the available data does not isolate the reasons for that variation. Electricity capacity alone should therefore not be used to infer water demand, and we draw no conclusion about which cooling technology should be preferred.

Existing frameworks do not resolve these inconsistencies. The reporting fields reviewed under NGER and the Safeguard Mechanism concern energy and emissions rather than water use,^{[1] [2] [3]} and our review did not identify an established NABERS Water rating for data centres or a generally applicable requirement for site-level water usage effectiveness reporting in the State and Territory instruments reviewed. **As noted at Annex F, the jurisdictional review was not comprehensive.**

The Committee does not need to settle a single long-run water forecast before requiring defined, site-level water reporting. A fixed water-volume threshold can identify facilities with high absolute withdrawal, if that is the policy objective. **What it cannot do is define a consistent class of facility by megawatts:** at 100 ML a year, the average IT load associated with that volume ranges from about 5 MW to about 571 MW across published WUE values †, so electricity capacity alone cannot determine whether a facility crosses a water-use threshold. (Arithmetic: Annex D.4.)

5. What the data gaps mean for competing forecasts

The forecasts before the Committee illustrate the practical consequences of the three data gaps. They cover different facility populations and use inconsistent assumptions about project development, utilisation, reporting boundaries and technology. **They should therefore not be treated as alternative estimates of one consistently defined quantity.**

Five categories of difference account for most of the divergence. (Each documented at Annex E.)

1. **Different facility populations.** The registers cannot produce an agreed list of facilities (section 2), so no two estimates are demonstrably counting the same population.
2. **Different project-realisation assumptions.** Forecasts apply different assumptions about duplicate requests, project likelihood and the proportion of proposed capacity that will proceed. One filing applies realisation factors by facility type — 68%, 55% and 56% — weighted by catchment mix;^[16] a network submission reports removing more than 3 GW of speculative demand, about 30% of one pipeline, at the application stage;^[17] and a national screening step removes 82% of requested capacity.^[15] **These percentages describe different analytical steps: the removal of duplicate or unlikely requests, the probability that a project proceeds, and the conversion of connection capacity into expected electricity demand. They are not directly comparable.**
3. **Different utilisation and ramp-up assumptions.** Even after a project is assumed to proceed, forecasts use different assumptions about utilisation, ramp-up and the relationship between connection capacity and actual electricity demand. The widely quoted "6 in every 7 MW is phantom demand" is a modelled estimate derived from screening, utilisation and absorption assumptions, rather than an observed measure of realised demand.^[15]
4. **Different modelling methods.** Some estimates are bottom-up project counts, while others blend project data with macroeconomic modelling — the 12 TWh FY30 headline is defined in its own source as $0.5 \times \text{project estimate} + 0.5 \times \text{economic estimate}$.^[14] **They are not alternative calculations of the same underlying quantity.**
5. **Different reporting boundaries and technology assumptions.** Electricity and water estimates also use different boundaries, denominators and technology assumptions. Nameplate capacity, IT load, whole-site load and actual electricity use are not interchangeable, while water demand depends on cooling architecture, climate and facility design (section 4).

Two measured series provide useful reference points. Statutory NGER returns for five corporations registered under the same name in both years rose from 1.99 TWh in 2023-24 to 2.36 TWh in 2024-25, **+18.8%**, against the 25.1% average annual growth assumed in the FY30 forecast.^[6] Separately, AEMO reports data centre load quarterly by region; in the quarterly reports reviewed, a separately identified data centre series first appears in 2026.^[19]

Neither series yet provides a stable facility-level baseline for validating a long-term forecast: the NGER data is corporate-level and incomplete, while the AEMO series is regional and has a short reporting history. (Rows, exclusions and an indicative reconciliation: Annex E.5.)

6. Recommendations

The three recommendations form a practical package. **Recommendation 1** would allow records about the same facility to be linked. **Recommendation 2** would show which obligations apply across jurisdictions and supply configurations. **Recommendation 3** would make published capacity, electricity and water figures more comparable.

Recommendation 1 — Identify the facility

This recommendation addresses the inability to determine whether records in different systems refer to the same physical facility (Gap A).

Establish, or publicly confirm, a persistent identifier for each facility. Use that identifier across NGER reporting, connection applications, State and Territory planning records, EPBC referrals and relevant water reporting.

Publish a crosswalk of non-sensitive fields so that public records relating to the same physical facility can be linked.

The identifier need not require a new central database. It could be added to existing records, supported by a public crosswalk containing only fields suitable for disclosure, with sensitive ownership, security and connection information remaining within government systems. Implementation would still require agreed field definitions, data-governance arrangements and intergovernmental coordination.

A minimum public retention period is needed alongside the identifier. Some development applications are no longer available through the relevant public online register five years after determination.^[21] Without a minimum retention period, the identifier could not support historical comparisons.

Recommendation 2 — Show who regulates it

This recommendation addresses uncertainty about how emerging obligations apply across NEM-connected, non-NEM and off-grid facilities (Gap B).

For each proposed or existing data centre obligation, **publish a coverage map** setting out how it applies to **NEM-connected facilities, facilities outside the NEM, and off-grid facilities**; and where no Commonwealth or NEM instrument applies, **identify any relevant State or Territory instrument and responsible agency. If no relevant instrument can be identified, the map should record a coverage gap.**

A published map would show in one place which obligations apply to each facility configuration and jurisdiction. **This would be more useful than case-by-case explanations that simply refer responsibility to a State or Territory without identifying the applicable instrument.** A coverage map is something the Commonwealth can deliver on its own initiative, and it would make gaps visible before new obligations are designed.

Public registers in different jurisdictions provide different parts of the required information: some identify the facility but not its capacity, some provide planning documents but no water data, and others do not classify the project as a data centre. **A map that named an agency for every cell would therefore misdescribe the position.** (Per-jurisdiction results, with method: Annex C.)

Recommendation 3 — Measure consistently

This recommendation addresses the inconsistent definitions that prevent capacity, electricity and water figures from being compared (Gap C).

Recommendation 3 has two elements: first, define the reporting boundary; second, specify the fields that must be reported.

Capacity figures should state whether they refer to IT load or whole-site load, whether they are designed or measured values, and whether the capacity is built, contracted or proposed. Efficiency metrics should state their denominator and measurement boundary.

Above a clearly defined threshold, require site-level reporting of water withdrawal, consumptive use, discharge volume and destination, source type, and WUE. WUE alone is not sufficient: an efficiency ratio does not measure a total.

The national data standard should also address the register problems identified at Annex C. Facility classifications should be searchable, and published capacity and water figures should state their reporting boundary.

The immediate problem is inconsistent definitions. Questions about reporting behaviour arise only after the quantities themselves are defined consistently. Undefined boundaries already produce incompatible numbers: the same 773 ML is published as consumption and as withdrawal in reports issued together (section 4), and at framework level, electricity used by customers' IT equipment in a colocation facility falls in the operator's scope 3 under the GHG Protocol and in the operator's scope 2 under NGER. ^[26] ^[2] (Annex D.)

Existing Australian frameworks already provide much of the required terminology. NABERS separates data centre efficiency into IT Equipment, Infrastructure and Whole Facility scopes, and the network forecasting material filed with the AER uses the maturity categories *Speculative*, *Enquiry to offer* and *Committed*. ^[16] Recommendation 3 would require existing classifications to be used consistently; it does not propose a new taxonomy.

Related matters for clarification

Annex A.1 identifies a statutory question about whether the publication condition in the *NGER Act* aligns with all of the corporate registration thresholds. Annex A.3 identifies an existing Commonwealth procurement mechanism relevant to government arrangements with data centre providers. **Neither point forms an additional recommendation.**

Implementation and scope

Implementation would combine direct Commonwealth action with intergovernmental coordination. The Australian Government could amend Commonwealth reporting arrangements, publish a national coverage map, set procurement conditions and pursue common data standards with the States and Territories. **Changes to State and Territory planning or water registers would require agreement with the relevant jurisdictions.**

The recommendations concern the structure, accessibility and comparability of public records. They do not prescribe a forecasting method or determine the level of any efficiency, water or emissions standard. Historical project outcomes and forecast assumptions would still need to be published and assessed separately.

Limitations

Evidence base. This submission relies on legislation, public registers, government documents and published company reports, referenced at Annex F; our own arithmetic is marked †. **Statutory interpretations are presented as questions for departmental confirmation. Nothing in this submission constitutes legal or financial product advice, or a recommendation concerning any company or financial product discussed.** Detailed limitations are set out at Annex F.2.

Technical annexes

Supporting evidence and methodology. Each annex sets out the documents relied on, the method used, and what the evidence does and does not establish.

Annex A	Instruments, thresholds, and the publication condition
Annex B	NEM, non-NEM and off-grid coverage
Annex C	Testing whether public registers can identify data centre facilities
Annex D	Why published water metrics are not directly comparable
Annex E	Why published forecasts cannot be compared directly
Annex F	References, and limitations of the evidence base

Terms used

Facility means the regulated unit as defined in the relevant instrument; *site-level* means that information is reported separately for each facility. *Whole-site load* means electricity drawn by the whole facility; *IT load* means electricity drawn by IT equipment. *Report* means submission to a regulator; *publish* means release to the public. *Withdrawal* means water taken; *consumption* means water not returned.

Annex A — Instruments, thresholds, and the publication condition

Several existing and proposed national frameworks are relevant to data centres, but they collect different information and apply at different thresholds. NGER records energy and emissions; the Safeguard Mechanism addresses covered direct emissions; the proposed AEMC measures concern connection standards and energy obligations; NABERS reports efficiency for participating facilities; and the National Data Centre Expectations are voluntary. **These frameworks should not be treated as alternative ways of measuring the same thing.**

The relevant policy question is whether they produce public, facility-level information that can be linked and compared. The table below identifies what each framework measures, when it applies and its current legal status. Sections A.1 to A.3 then examine three specific issues: whether NGER registration necessarily leads to publication, what the public NABERS register provides, and how procurement can use NABERS ratings.

Table A.1 — What each framework measures, when it applies and its legal status

Framework	What it measures	Threshold as written	Illustrative continuous-load equivalent	Legal status
NGER (facility) ^[1]	Energy produced, energy consumed, and emissions	100 TJ energy, or 25 kt CO ₂ -e	about 3.2 MW of continuous consumption ⁺	In force
NGER (corporate group) ^[1]	The same, totalled to the company	200 TJ, or 50 kt	about 6.3 MW ⁺	In force
Safeguard Mechanism ^[3]	Scope 1 (covered) emissions	100,000 t per facility	No equivalent can be stated — see note	In force
AEMC access standards (Package 2) ^[5]	Import capacity; inverter behaviour	30 MW; 100 MW (Tier 3)	30 / 100 MW of import capacity	Draft. Final determination due late October 2026
AEMC data centre energy obligation ^[4]	Added renewable supply and firm capacity for purchased power	Not yet set	Not yet set	Advice. Considered by Ministers 28 July 2026
NABERS Energy ^[8]	Power Usage Effectiveness (a ratio)	Not triggered by facility size	Not applicable	Generally voluntary. Can be a procurement or contract condition
National Data Centre Expectations ^[25]	General expectations	—	—	Voluntary (23 March 2026)

The megawatt equivalents in the fourth column are arithmetic illustrations only. **They are not statutory thresholds.** ⁺ Our arithmetic: 100 TJ ÷ 3.6 ÷ 8,760 = 3.17 MW; 200 TJ likewise = 6.34 MW. **No megawatt equivalent is given for the Safeguard threshold.** That threshold applies to a facility's actual covered emissions, which depend on fuel, plant efficiency and operating hours. The 0.539 t CO₂-e per MWh figure at Schedule 1 s 57(3) of the Safeguard Rule is a default emissions intensity used to set a baseline for the electricity generation production variable, not an emissions factor for determining whether a facility reaches 100,000 t, and dividing one by the other would produce a figure with no statutory meaning.

The table shows why the thresholds cannot be substituted for one another. For example, a grid-supplied facility operating continuously at approximately 25 MW would generally exceed the NGER energy threshold, remain below the proposed 30 MW access-standard threshold and, in the absence of substantial on-site fossil generation, would not ordinarily incur a direct Safeguard liability. **The three outcomes arise because the frameworks measure different quantities.** Whether NABERS applies does not depend on size at all.

A.1 NGER registration does not necessarily make reported totals public

NGER registration and public disclosure are separate questions. A corporation may be required to register and report because it reaches an energy-consumption threshold, while publication under section 24(1B)(a) is linked to an emissions threshold.^[1] The issue is therefore whether a large electricity user with relatively low direct emissions may report data that is not published through this particular mechanism.

Under section 24(1B)(a), publication of a corporation's reported totals is conditional on the corporate group meeting the emissions threshold in section 13(1)(a) — 50,000 t CO₂-e. Registration is separately triggered by section 13(1)(c) — 200 TJ consumed, an energy threshold.^[1] **On our reading, meeting only the energy-registration threshold does not make a corporation's totals eligible for publication through this particular mechanism.**

The Committee may wish to ask the responsible department whether the publication condition in section 24(1B)(a) is intended to align with all registration thresholds in section 13, including the energy-consumption threshold.

An illustrative calculation is provided only to show the possible scale of the difference between the energy and emissions thresholds. It does not establish that any particular corporation's data is withheld. The lower edge is 200 TJ of energy consumed; the upper edge is 50,000 t CO₂-e converted to electricity at an assumed grid emissions intensity — about 257 TJ + at roughly 0.70 t/MWh, or about 450 TJ + at roughly 0.40 t/MWh. Over 8,760 hours that is roughly 6.3 MW up to somewhere between 8.2 MW and 14.3 MW +. Whether any given corporation falls in that range also depends on its scope 1 emissions, its reporting boundary, how section 13 is applied in practice, and whether other publication powers reach the same data — questions we have not examined. Sections 25 (commercially sensitive information) and 24(1C) (publication in ranges) would need to be considered alongside section 24(1B). This is a reading of the provisions offered for departmental confirmation.

A.2 NABERS publishes detailed data for rated facilities, but coverage is limited

NABERS is sometimes described only as an efficiency-rating system. The public register shows that, for participating data centres, it may also publish absolute annual electricity use alongside PUE. **The relevant limitation is not that the metric produces no usable data, but that only a small and inconsistently defined part of the sector appears in the register.**

We downloaded the full public NABERS rating register on 15 August 2026 (4,658 current ratings).^[8] **Eleven** Australian data centres currently hold a NABERS Energy rating: eight in New South Wales, two in Victoria and one in Western Australia. Ten are rated on the Infrastructure scope and one on Whole Facility. Across the ten Infrastructure-scope ratings, measured PUE runs 1.23 to 1.80, median 1.54, and the register publishes absolute annual electricity per named premises, about 226 GWh in total.

NABERS therefore demonstrates that facility-level electricity and efficiency data can be published. **Its present limitation for sector-wide analysis is coverage: the register includes only rated facilities, most of which use the Infrastructure rather than the Whole Facility scope.**^[29]

Published estimates of the national facility population range from roughly 160 to 300, but those estimates count different units — independent facilities, campuses, data halls, colocation sites and enterprise server rooms — and therefore do not support a reliable coverage percentage.

One Whole Facility entry reports a PUE of 0.40 without a corresponding energy figure. Because PUE is defined as total facility energy divided by IT equipment energy,^[29] a value below 1.0 is inconsistent with the metric's definition. That entry is excluded from the range reported above.

Because ten of the eleven are rated on the Infrastructure scope, which excludes IT equipment load, the public register does not generally allow whole-site electricity to be derived from an Infrastructure rating. A Commitment Agreement lets a proposed facility publish a future rating *target*; a target is not a verified operating result, and it is a ratio in any case.

A.3 Government procurement can use an existing rating framework

Term of reference (a) also concerns arrangements between the Australian Government and AI companies. Existing Commonwealth procurement settings provide one mechanism through which data centre performance requirements can be applied without creating a new regulatory metric.

The National Data Centre Expectations allow a 5-star NABERS Energy rating, or an accepted equivalent, to become a condition of supplying data centre services to government.^[25] **The Committee may wish to establish how often this condition is used, how an "accepted equivalent" is defined and whether the resulting performance information is published.**

Annex B — NEM, non-NEM and off-grid coverage

Whether a regulatory obligation applies to a data centre depends on two separate questions: where and how the facility is connected, and whether it reaches the relevant statutory threshold. NGER and the Safeguard Mechanism apply nationally but are triggered by actual annual energy use or emissions. The proposed access standards depend on connection capacity and apply through the National Electricity Rules, while the geographic reach and thresholds of the proposed additionality and firm-capacity obligations are not yet settled. Table B.1 sets out these legal triggers and geographic limits. Table B.2 applies them to common facility configurations.

Three concepts should be kept separate. **"Outside the NEM"** describes geographic and legal coverage. **"Off-grid"** describes a facility with no grid supply connection. **"Self-supplied"** describes a facility that meets its principal demand from on-site generation but may retain or propose a connection for export, backup or other purposes. A facility may fall into more than one of these categories.

B.1 How each framework is triggered and where it applies

The proposed access standards and the proposed data centre energy obligations are at different stages and may have different geographic reach. The access standards would operate through the National Electricity Rules and therefore would not apply in the Northern Territory or Western Australia. The implementation route for the proposed additionality and firm-capacity obligations has not yet been settled.

Table B.1 — Trigger and geographic reach of each framework

Instrument	What triggers it	Geographic reach
NGER reporting ^[1]	Actual annual energy produced, energy consumed, or emissions, assessed separately against each threshold	National
Safeguard Mechanism ^[3]	Actual covered (scope 1) emissions reaching 100,000 t	National
AEMC access standards (Package 2, draft) ^[5]	Connection import capacity and the technical characteristics of the load	National Electricity Market — the National Electricity Rules do not apply in the NT or WA
AEMC additionality and firm capacity (advice) ^[4]	Threshold not yet set	Not yet established in the published advice

B.2 Indicative application to common facility configurations

The table applies the published rules and proposals to illustrative facility configurations. It distinguishes between obligations that clearly apply, obligations that depend on an annual threshold being reached, and proposals whose final scope remains unsettled.

Table B.2 — Indicative application to common facility configurations

Configuration	NGER reporting	Safeguard obligation	Access standards (draft)	Additionality / firm capacity (advice)
Grid-supplied, in the NEM	If any annual threshold is reached	Only if covered emissions reach 100,000 t	By import capacity: 30 MW and 100 MW bands as proposed	Depends on final scope and threshold

Configuration	NGER reporting	Safeguard obligation	Access standards (draft)	Additionality / firm capacity (advice)
Grid-supplied with on-site solar	As above, and generation counts toward the energy-production threshold	Only if covered emissions reach 100,000 t	By import capacity	Depends on final scope and threshold
Grid plus on-site fossil generation	If any annual threshold is reached	Depends on actual covered emissions, not on megawatts	By import capacity	Depends on final scope and threshold
Off-grid, fossil-fired	If any annual threshold is reached	Depends on actual covered emissions, not on megawatts	None — no connection	Depends on final scope; the advice describes grid-connected facilities
Off-grid, fully renewable	If any annual threshold is reached; below all of them, none	None	None — no connection	Depends on final scope; the advice describes grid-connected facilities
Grid-connected or self-supplied facility in the NT or WA	Unchanged — NGER applies nationally if an annual threshold is reached	Unchanged — the Safeguard Mechanism applies nationally if covered emissions reach 100,000 t	None under the proposed NEM rules — the National Electricity Rules do not apply in the NT or WA	Not yet established — the implementation route and geographic scope remain unsettled

In this table, "None" means that the particular energy instrument does not apply in the stated configuration. Other planning, environmental, water, electricity and contractual requirements may still apply. **"Self-supplied" does not necessarily mean physically disconnected from every electricity network.** A facility may generate the electricity it consumes while retaining or proposing a connection through which surplus generation could be exported.

Two features of the table require explanation. First, NGER thresholds are based on annual energy and emissions quantities, not connection capacity. A facility with a large connection may therefore remain below an annual threshold if it operates at low utilisation or is commissioned part-way through a reporting year. Second, an off-grid renewable facility must assess the NGER energy-production and energy-consumption thresholds separately (regs 2.25, 2.26, 4.19, 4.20).^[2] If none of the relevant annual thresholds is reached, no NGER report is required. Under reg 5.03, net energy consumption is consumption less the energy commodities the facility itself produces, so a facility's published scope 1, scope 2 and net energy figures may all be close to zero even where it does report.

B.3 What the AEMC advice says about grid connection and NEM coverage

At page 17, the AEMC advice describes the proposed data centre energy obligations as potentially applying to data centres "connected to the grid" above a threshold size. At page 47, it states that the accompanying network cost-recovery reform "would only apply to the NEM".^[4] Because the two statements concern different elements of the proposed package, the second should not be treated as defining the geographic reach of every proposal in the advice.

"These policies could be applied to all types of data centres **connected to the grid** that are above a certain threshold size and implemented in stages." (p 17)

"This would **only apply to the NEM**." (p 47)

The advice does not separately explain how the proposed obligations would apply to off-grid or islanded facilities. Its references to behind-the-meter assets concern grid-connected facilities using on-site assets to manage their own obligations.^[4]

B.4 Two Northern Territory proposals illustrate different non-NEM supply configurations

Project Ares and Beetaloo Digital are included because both are large proposed data centre developments outside the NEM, but their published supply arrangements are not identical. Project Ares is expressly described as entirely off-grid. Beetaloo Digital is presented as meeting its own demand from on-site generation without drawing from the existing Territory grid, and as adding new generation capacity to the Territory.

The comparison demonstrates why a national coverage map should record both jurisdiction and supply configuration. **Describing both projects simply as "off-grid" would obscure a potentially relevant distinction between a facility with no network supply and a self-supplied facility that may nevertheless contemplate supplying electricity to the network.** For both, NGER may still apply if the relevant annual thresholds are reached, while the proposed NEM access standards would not.

Project Ares: published project characteristics	
Status	Referred. EPBC 2026/10560, referral valid 29 June 2026; <i>Awaiting additional information</i>
Location	Murrانji Station, Barkly region, about 683 km south of Darwin
Stated IT capacity	1 GW, in two stages
Stated generation	3,000 MWp solar, 16 GWh battery, 900–1,100 MW gas
Stated water	4 GL a year, groundwater, semi-arid region
Relationship to the NEM	Outside the NEM, and the referral states the facility runs off-grid and draws no power from the NT network

Sources: EPBC Act Public Portal, referral 2026/10560 and supporting documents;^[7] the proponent's public announcements. All quantities in the table are taken from the proponent's published materials; Heliovolcan has not independently estimated them.

Beetaloo Digital: published project characteristics	
Status	Proposed; feasibility work and regulatory approvals remain to be completed. The Northern Territory Government has granted a 12-month exclusive commitment over a 185-hectare site
Location	Weddell, about 30 km south of Darwin
Stated scale	Up to two hyperscale data centre campuses
Stated generation	Up to 2 GW of on-site gas-fired generation, supplied with gas from the Beetaloo Basin
Stated water	Not stated in the cited Northern Territory Government announcement
Relationship to the NEM	Outside the NEM
Relationship to the NT grid	The project is described as designed "to add new generation capacity to the Territory rather than draw from the existing grid", and as a potential source of large-scale baseload power. The final connection arrangement has not been publicly established

Source: Northern Territory Coordinator, "Beetaloo Digital advances plans for Weddell data centre", 5 August 2026.^[30] The figures and the proposed supply arrangement are statements about a project at an early development stage and have not been independently verified.

B.5 How Project Ares' on-site solar would be treated under NGER

Of the two proposals, Project Ares also raises a separate and documentable NGER reporting question, because its EPBC materials specify both large-scale on-site solar generation and data centre load. Generation from on-site solar is reported to the regulator under the NGER Regulations 2008 — reg 4.20(1)(c) lists solar generation, reg 4.20(2)(a) requires the portion used by the facility to be identified separately, and it counts toward the NGER energy-production threshold.^[2] On-site solar counts towards the NGER energy-production threshold, but it does not change the import-capacity classification used by the proposed NEM access standards. The public NGER data also does not identify on-site solar generation as a standalone facility-level sustainability measure.^[2]

Annex C — Testing whether public registers can identify data centre facilities

To test whether existing public records could support Recommendations 1 and 3, we attempted to compile a facility-level dataset from State and Territory planning, environmental and water registers. For each facility, we looked for three types of information: a record that identifies the physical project, a stated electricity-capacity figure and a reported water-use figure.

The review covered public registers in New South Wales, the Australian Capital Territory, South Australia, Victoria and Queensland. Where machine-readable records were available, we searched project descriptions and classifications for data-centre-related terms. We also attempted to match known facilities to planning records using applicant names, operator names, addresses, lot and plan descriptions, and stated capacity. The registers and record counts are identified in references ^[18] and ^[20]–^[24].

None of the jurisdictions reviewed provided all three types of information consistently. Some registers omitted projects below a State-level assessment threshold; some did not classify the proposed use as a data centre; some did not publish capacity or water fields; and some ceased to provide access to documents after a defined period. Table C.1 shows where the public record became incomplete in each jurisdiction.

Table C.1 — Information recoverable from the public registers reviewed

Jurisdiction	Information available	Information missing or inaccessible
NSW	The most complete range of information of the jurisdictions tested: capacity appears in official descriptions and in EIS documents, and one facility publishes annual water. It still does not answer all three questions for all facilities	Projects below the State Significant Development threshold do not enter the State register at all; they would have to be found through individual council portals, which have no common entry point or classification. Some known facilities could not be matched to a case in the State-level registers searched
ACT	The register reliably identifies projects described as data centres, and was the most searchable of those tested	The documents returned in our search did not provide electricity-capacity figures. Records are available only from 2019, and determined applications cease to be accessible through the online register after five years, so the original applications for several existing Canberra facilities can no longer be obtained through it. None of the 41 environmental-impact records returned in the separate search was classified as a data centre
SA	The keyword search of 207,770 machine-readable records returned seven results and identified four facilities not found in the published facility lists, including three containerised installations within solar farms	No capacity field and no water field

Jurisdiction	Information available	Information missing or inaccessible
VIC	Section 122ZJ of the <i>Water Act 1989</i> requires water corporations to publish the names of non-residential customers using at least 100 ML a year — 80 named customers across three utilities in FY2024-25. ^[18] Victoria's water connection application form is the only one found that asks for both staged MW and WUE	The Ministerial permits register provides limited information for matching facilities because it does not publish the applicant's name. Several operating facilities could not be matched to a planning permit
QLD	Documents downloadable in bulk	Not searchable by use. A full-text search of all 1,703 decided applications returned no occurrence of "data cent", and the same search of 123 Coordinated Projects returned none

The Queensland result requires careful interpretation. The search returned no occurrence of "data cent", but this does not show that no data centre has undergone State assessment. It shows that a keyword search cannot identify projects lodged under another use description.

The Western Australian example demonstrates this classification problem. Application DAP/25/02926 is described in the panel agenda as *Proposed Warehouses and Ancillary Structures* at Lots 1, 10 and 1 (Nos 307, 323 and 329) Kenwick Road and Lot 500 (No 582) Bickley Road, Maddington, while material in the same published document set describes the project as a data centre campus with "150MW of IT Capacity".^[27] A keyword search based only on the formal development description would not identify it as a data centre. The point concerns the register's classification vocabulary, not the conduct of the applicant or the panel.

Taken together, the results show that the registers reviewed cannot produce a consistent facility-level dataset. A persistent identifier would improve matching across systems, while common capacity and water fields would address the missing information shown in Table C.1.

Annex D — Why published water metrics are not directly comparable

Published estimates of data centre water use cannot be compared until three questions are answered: what water quantity is being reported, what facility or organisational boundary it covers, and what denominator is used. This annex addresses those questions in sequence. Section D.1 examines the difference between withdrawal and consumption. Section D.2 compares the reporting boundaries used in published Australian WUE figures. Section D.3 considers what regional PUE and WUE data can and cannot show. Section D.4 applies those findings to the design of a water-reporting threshold.

D.1 The same quantity is reported under two different definitions

NEXTDC Limited's FY25 reports, issued together, record the identical portfolio water quantity under two different definitions:^{[10] [11]}

Document	Where	Wording as published	Figure
NEXTDC FY25 ESG Report	p 29, narrative	"consumed approximately 773 ML of water across our data centre portfolio"	773 ML, consumption
NEXTDC FY25 ESG Report	p 32, metrics table	"Total water consumption — Potable Water (municipal water supply)"	773.31 ML, consumption
NEXTDC FY25 Climate and Nature Report	p 23, metrics table	"Water withdrawal — Potable Water"	773 ML, withdrawal

At evaporatively cooled facilities, withdrawal and consumption differ materially. The reports appear to use "consumption" and "withdrawal" for the same portfolio-level quantity. This submission does not determine how the 773 ML figure should be classified and makes no finding about the reporting process that produced the two labels. **Because the two terms describe different water measures, the published figure cannot be interpreted consistently or compared reliably with figures reported by other operators.** The example supports standardising reporting definitions, rather than simply requiring more disclosure.

Standardising the water quantity is only the first step. Even where the same metric is used, published figures may still describe different facilities, regions or corporate portfolios. Section D.2 examines that second source of non-comparability.

D.2 Australian WUE figures use different reporting boundaries

Published WUE values do not all describe the same reporting population. The examples below include an operator's Melbourne facilities, an AWS region, NEXTDC's national portfolio and an attributed industry average. **A value reported for one of these populations cannot be compared directly with a value reported for another.**

Published WUE	Reporting boundary	Source
0.02 L/kWh (2024)	One operator's Melbourne facilities, stated as litres <i>withdrawn</i> per kWh of IT load, with 96% of the year on free-air cooling	Victorian Government <i>Industry Water Connection Guide</i> , p 7 ^[13]
0.02 L/kWh (2024)	One region — AWS Asia-Pacific (Melbourne), ap-southeast-4	AWS published regional PUE/WUE table ^[12]
2.26 (FY25)	One operator's national fleet — NEXTDC	NEXTDC FY25 Annual Report, p 16 ^[9]
1.8	Stated Australian industry average	Cited in the Victorian guide, attributed there to Mandala (2025) ^[13]

The first two entries appear to reproduce the same Melbourne regional figure. The Victorian guide reports 0.02 L/kWh "across its Melbourne data centres", while the AWS table reports 0.02 L/kWh for the Melbourne region. **They should therefore be treated as one published value appearing in two sources, not as two independent observations.** The industry average of 1.8 is likewise not an independent measurement: the guide attributes it to a consultancy report.

Each figure accurately reflects what its source reports, but at a different organisational or geographic level. **The figures cannot be compared directly because they describe different reporting populations.** Because the sources do not publish their methods on a common basis, the values are reported here as stated and are not used to calculate comparisons between operators.

NEXTDC's FY25 reports give the same metric as 2.26 in the Annual Report and 2.25 in the ESG and Climate and Nature Reports.^{[9] [10] [11]} That difference is consistent with rounding and is not of the same kind as the withdrawal/consumption question at D.1.

Reporting boundary explains why the four headline values cannot be ranked against one another. A separate question is whether regional PUE and WUE data reveal a consistent relationship between electricity efficiency and water use. Section D.3 considers that question.

D.3 Published regional PUE values do not provide a reliable proxy for WUE

AWS publishes both PUE and WUE for several regions. The figures allow a limited check of whether similar electricity-efficiency values correspond to similar water-efficiency values.

AWS region	PUE	WUE (L/kWh)
Asia-Pacific (Melbourne)	1.07	0.02
Asia-Pacific (Sydney)	1.16	0.12

AWS region	PUE	WUE (L/kWh)
Europe (Ireland)	1.11	0.03
U.S. East (Northern Virginia)	1.15	0.12
Asia-Pacific (Singapore)	1.32	1.68
Asia-Pacific (Jakarta)	1.40	2.75
Global	1.15	0.15

Source: AWS Australia fact sheet and published regional PUE/WUE table, 2024 figures.^[12]

Regions with similar reported PUE can have different reported WUE. Sydney and Northern Virginia both report WUE of 0.12 with PUE values of 1.16 and 1.15, while Ireland reports a similar PUE of 1.11 but a lower WUE of 0.03. Melbourne reports both the lowest PUE and the lowest WUE in the table, whereas Singapore and Jakarta report higher values for both metrics.^[12]

This small regional table does not isolate the effects of climate, cooling systems, utilisation, facility design or fleet composition. It therefore cannot support a universal conversion between PUE and WUE. **The relevant conclusion for reporting design is narrower: water use should be measured directly rather than inferred from an electricity-efficiency metric.** It follows that a single long-run water forecast need not be settled before defined site-level reporting is required.

D.4 A 100 ML reporting threshold corresponds to different average IT loads at different WUE values

The preceding sections show that WUE varies across published reporting populations and cannot be inferred reliably from PUE. This has a direct implication for threshold design. Recommendation 3 proposes site-level water reporting above a clearly defined threshold. If annual water reporting were triggered at 100 ML, the average IT load associated with that water volume would depend on the facility's WUE.

Published WUE (L/kWh)	Average IT load corresponding to 100 ML/year †
2.26 — NEXTDC national portfolio ^[9]	approximately 5.1 MW
0.12 — AWS Sydney region ^[12]	approximately 95 MW
0.02 — AWS Melbourne region ^[12]	approximately 571 MW

† Illustrative arithmetic based on continuous average IT energy use over 8,760 hours. The calculation converts an annual water volume into average IT load; it does not estimate nameplate capacity or design capacity, and it does not attribute a particular WUE to a cooling technology.

The same 100 ML threshold can therefore apply to facilities with very different average IT loads. A water-volume threshold identifies absolute water use; it does not define a consistent class of facility by MW. **Water-reporting requirements should accordingly be based on the relevant water quantity and reporting boundary, rather than on an assumed conversion from electricity capacity.**

Victoria's statutory disclosures identify non-residential customers using at least 100 ML a year, rather than necessarily reporting individual facilities on a common site-level boundary.^[18] The absence of an identifiable data centre name from those lists could mean that a facility falls below the threshold, that several sites are aggregated under a customer name, or that the customer is not readily identifiable as a data centre operator. **It does not establish low sector-wide water use.**

Annex E — Why published data centre forecasts cannot be compared directly

The estimates before the Committee reflect two related but different policy concerns. The first is how much electricity future data centres may consume, which affects generation and network planning. The second is how much of the capacity currently appearing in connection pipelines is likely to become operating demand, given duplicate, speculative or immature projects.

Two Oxford Economics reports sit near the centre of that debate. The July 2025 report prepared for AEMO forecasts future data centre electricity consumption for system planning. The November 2025 briefing commissioned by Amazon Web Services argues that connection-request totals substantially overstate the demand likely to materialise. The reports are particularly useful to compare because they were produced by the same consultancy but for different commissioners, purposes and analytical questions.

Neither report directly observes how a historical cohort of data centre projects progressed from connection request to operating load. Their headline results depend on choices about the project population, screening, utilisation, ramp-up and mature demand. The sections below identify those choices before considering whether measured NGER and AEMO data can provide an external check.

E.1 The two reports answer different policy questions

	July 2025 ^[14]	November 2025 ^[15]
Prepared for	AEMO	Amazon Web Services
Purpose	Forecast consumption for system planning	Estimate how connection requests translate into expected electricity demand
Headline	12.0 TWh FY30; 34.5 TWh FY50	"6 in every 7 MW is phantom demand"

The AEMO report estimates future electricity consumption. The AWS-commissioned briefing estimates how connection-request capacity may reduce to actual grid demand. The first is a consumption forecast; the second is a pipeline-adjustment analysis. **Their headline figures are therefore not directly comparable.**

E.2 AEMO's 12 TWh forecast averages two different modelling approaches

The AEMO-commissioned report calculates its FY2030 estimate by giving equal weight to a project-by-project model and a macroeconomic model adjusted for PUE, server utilisation and Koomey's law (§6.3, p 47).^[14] **The published 12 TWh figure is therefore an average of two model outputs, rather than a direct count of identified projects.**

Each half of that average moves for different reasons. The project-by-project half changes with the projects included and the assumed timing of their development. The macroeconomic half changes with the economic model and the efficiency assumptions. The published total also changes with the decision to weight the two methods equally. A forecast that differs from this one may therefore differ because of method, rather than because of a different view of the same project pipeline.

The reports also use "load factor" and "utilisation" differently. The values 51%, 47% and 45% are labelled "IEA load factor" in one report and "utilisation" in the other, while 0.75, 0.85 and 0.80 are labelled "AEMO load factor" in one report and "load factor" in the other.^{[14] [15]} Any comparison should therefore rely on the definitions and calculation steps in each report, rather than on the labels alone.

E.3 The AWS-commissioned briefing applies three reductions to connection-request capacity

The November 2025 briefing begins with 44 GW of connection requests and applies three main adjustments: project screening, utilisation and absorption. It arrives at 2.8 GW of expected demand at maturity.^[15]

Step	Capacity	Effect
Connection requests (2025 planning inputs)	44 GW	—
After screening for likelihood and duplicates	7.9 GW	removes 82% of the requested capacity
If every screened project proceeds, at 48% utilisation	3.8 GW	utilisation assumption
Expected actual draw at maturity	2.8 GW	absorption assumption

The 2.8 GW result is a modelled scenario. Most of the reduction occurs in the first step, when screening reduces the connection pipeline from 44 GW to 7.9 GW — about 82% of the requested capacity. The remaining reduction depends on assumed utilisation and absorption. We could not find published evidence for the absorption assumption as applied to existing facilities.

The headline "6 in every 7 MW" implies that 14.3% of requested capacity becomes demand. The separate statement that 7 MW out of every 100 MW becomes actual draw implies 7%; that statement is made about the connection requests AEMO received for the 2025 IASR and is expressed as draw "in 33 years" (pp 1 and 4).^[15] Because the two ratios use different descriptions and different bases and time horizons, neither should be cited without the underlying calculation. **The more transparent account is the published sequence from 44 GW to 7.9 GW, 3.8 GW and finally 2.8 GW.**

The Oxford briefing is not the only source that adjusts connection-request capacity. Network businesses also apply project-realisation, ramp-up and load-factor assumptions when preparing regulatory forecasts. The following examples show that these adjustments occur at different stages and do not represent a common retention measure.

E.4 Network forecasts adjust proposed capacity at different stages and using different methods

Network forecasts do not apply a single standard "retention rate" to connection requests. The documents reviewed make adjustments at different stages of the project pipeline and for different purposes.

The forecast prepared for CitiPower, Powercor and United Energy and filed with the AER does not apply a single retention rate. Its assumptions table (p 28) records **mature load realisation factors of 68% for hyperscalers, 55% for colocators and 56% for edge, telco and other**, weighted by the mix of data centre types in the network's catchment in FY31, and then applies a separate ramp factor and a separate load factor on top.^[16] The document therefore does not provide a single overall project-realisation rate.

A joint submission by the NSW distribution businesses records a different kind of adjustment again — not a forward realisation rate but a removal already made. It states that "more than 3 GW of speculative demand has been removed from the application stage of Ausgrid's connections pipeline – approximately 30 per cent of the total pipeline" (p 8).^[17]

A facility-type realisation factor, the removal of applications already judged speculative, and a national screening adjustment are not equivalent measures. **They cannot be compared as though each represents the proportion of requested capacity ultimately built and operating.** Network forecasts combine several separate assumptions, including project realisation, construction ramp-up, utilisation and mature load. Because the reviewed documents define and apply these assumptions differently, their adjustment percentages cannot be compared directly.

Public historical project cohorts would allow these assumptions to be tested against realised outcomes, but the cohorts underlying the reviewed analyses are not published: the Oxford Economics briefing states that its cohort is not published, while the network analysis presents its pipeline under anonymous identifiers. **The Committee may therefore wish to ask AEMO and the network businesses:**

1. For historical connection cohorts, what proportion of requested capacity ultimately became operational demand?
2. Have the published project-realisation, ramp-up and load-factor assumptions been tested against those outcomes?

E.5 NGER corporate data provides a partial measured baseline

The forecast analyses above are based on project pipelines and model assumptions. NGER provides a partial empirical comparison because it records annual energy use reported under legislation. However, the regulator publishes the data by **registered corporation** rather than by physical facility.^[6] We constructed a two-year comparison for five corporations whose registered names were unchanged across both years. A sixth facility is shown separately because its ownership and registered corporation changed between years.

Reporting corporation	2023-24 net energy (GJ)	2024-25 (GJ)	Change
AirTrunk	2,275,838	2,886,668	+26.8%
NEXTDC	1,744,228	1,971,147	+13.0%
CDC	1,367,770	1,729,124	+26.4%
Equinix	1,449,889	1,557,358	+7.4%
Digital Australia Investment Management	322,737	365,281	+13.2%
Five corporations registered under the same name in both years	7,160,462 (1.99 TWh)	8,509,578 (2.36 TWh)	+18.8% [†]
Global Switch Australia → HMC Digital Infrastructure	422,466	443,481	<i>matched by us across an ownership change</i>
With that pair matched	7,582,928 (2.11 TWh)	8,953,059 (2.49 TWh)	+18.1% [†]

Corporations included in the comparison

- Amazon Corporate Services was excluded because its NGER registration covers activities beyond data centre operations, so including it would present a mixed-business total as a data centre figure.
- Global Switch Australia and HMC Digital Infrastructure were matched manually across the ownership change described at section 2, and are shown separately from the unchanged-name series. That manual matching is exactly the step a public facility identifier would remove.
- The resulting figures represent selected corporations associated with data centre activity, not the entire Australian data centre sector.

Comparison with forecast growth. The five-corporation series increased by 18.8% between 2023-24 and 2024-25, or 18.1% including the matched pair, while the FY30 forecast assumes average annual growth of 25.1%.^[6]^[14] These figures are of a similar order, but they are not directly comparable because the corporate population, reporting boundaries and time periods differ. **Two annual observations provide a preliminary reference point, not an established trend or a validation of the forecast.**

The 2.49 TWh recorded for the expanded series is approximately **64%** ⁺ of the 3.9 TWh attributed to FY2025 by the sector model. This is only an order-of-magnitude comparison: the denominator is modelled, the corporate population is incomplete, and the reporting boundaries differ. **It should not be interpreted as a coverage rate.**

The comparison shows that NGER records at least 2.49 TWh for the selected corporations on the basis used here. **It does not establish the share of total sector electricity captured, because the corporate population is incomplete and the modelled denominator uses different boundaries.** A disclosure obligation would principally add facility-level detail, comparability and water-use reporting.

The same model reports 3.9 TWh of FY2025 consumption and forecasts that the "existing stock of data centres is expected to consume 8.5 TWh of energy by FY30", growing at 7.8% a year until FY41, "when all existing data centres have reached their mature load" (p 16).^[14] That comparison cannot be interpreted without definitions of "existing", built, commissioned, fitted-out and available capacity, and we have not obtained them. It is recorded here as a question for clarification, not as evidence supporting a recommendation.

E.6 AEMO now provides measured regional data centre demand

AEMO's *Quarterly Energy Dynamics* provides a second empirical reference point because it reports measured regional demand rather than project announcements or connection requests. In Q2 2026, average data centre demand was **451 MW** in New South Wales, approximately 35% higher than a year earlier. In Victoria, it increased from 104 MW in Q2 2025 to **204 MW** in Q2 2026 (p 9).^[19]

The series can be used to monitor near-term demand growth in the two reported regions. **It cannot yet provide a national or facility-level baseline**, because other jurisdictions are not separately identified and the published history is short.

Annex F — References, and limitations of the evidence base

Every numbered marker in the text — for example ^[1] — links to the corresponding entry below. Quantities marked ⁺ are our own arithmetic on the published figures.

F.1 References

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F.2 Limitations of the evidence base

Jurisdictional coverage. The register review was not systematic across all eight jurisdictions. Tasmania and Western Australia were not reviewed to the same depth as New South Wales, the ACT, South Australia, Victoria and Queensland. No conclusion in this submission treats an incompletely reviewed jurisdiction as one in which no relevant records exist.

Availability of records. Where a register no longer returns a record, we say only that the record is not available through that channel. We do not know whether it survives in departmental archives or is obtainable under freedom-of-information processes, and we make no claim that any record has been destroyed. Helio Vulcan's internal facility register is not reproduced because it contains proprietary matching work and remains subject to revision as new attribution evidence becomes available; our own planning roster holds only two determinations before 2018. Of the 62 NSW facilities in that internal comparison list as at August 2026, 19 could not be matched to a case in the State-level registers using facility name, operator name, address, or lot and plan; because the list is proprietary and subject to attribution changes, that count is not a population estimate.

What we could not obtain. Historical connection cohorts are not public, so the published capacity-adjustment assumptions at Annex E.4 cannot be independently tested against historical outcomes using public information. We also did not identify published definitions for an "accepted equivalent" procurement rating, a timetable for a NABERS data centre water framework, or a complete correspondence between NABERS star ratings and PUE values. Reference [12] was publicly accessible when retrieved on 16 August 2026; a copy is retained by the author. Reference [28] cites the author-published copy available at the date of submission.

Interpretations and indicative reconciliations. The statutory interpretations at Annex A.1 and the discussion of on-site generation under regulation 2.16 are presented for departmental confirmation rather than as independent findings. Similarly, the 64% comparison and the existing-stock consumption comparison at Annex E.5 are indicative comparisons that use modelled denominators. None of these interpretations or comparisons is relied upon in support of the three recommendations.